

biosignals including electrocorticogram (ECoG), electromyogram (EMG), electrocardiogram (ECG), electrooculogram (EOG), electroencephalogram (EEG), respiration and many others. In addition, the interfacing toolboxes and drivers for Octave, MATLAB, Python, PHP, Perl, Ruby, Tcl, C and C++ are also available. The key areas of brain-computer interfaces, psychology, neuroinformatics, cardiovascular systems, neurophysiology and sleep research are effectively processed in BioSig.



Figure 1: Elements of bioinformatics

GenomeTools (<http://genometools.org/>)

GenomeTools is open source software for the analysis of genome and biological parameters. It has a free library of tools for bioinformatics. The APIs in C are available with detailed manuals. In addition, the deep analysis of biological structures is integrated in GenomeTools.

Working with BioPython for molecular biology (<http://biopython.org/>)

BioPython provides the set of tools and libraries for the analysis and computation of biological structures. It is available for free and open source distribution and is promoted by the Open Bioinformatics Foundation (OBF). It can parse the files of bioinformatics into the data structures that can be processed by Python code.

The following international formats are supported in BioPython:

- UniGene
- PubMed
- GenBank
- Medline
- GenBank
- FASTA
- Clustalw
- Blast

Use the following command to install BioPython in Ubuntu:

```
$ sudo apt-get install python-biopython
```

The following command will install the documentation along with BioPython:

```
$ sudo apt-get install python-biopython-doc
```

BioSQL (<http://biosql.org>) can be used with BioPython to store a biological database. To integrate BioSQL, the following instruction is executed:



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See also What's new.

Files

Biopython 1.70

- [biopython-1.70.tar.gz](#) 15Mb - Source Tarball
- [biopython-1.70.zip](#) 16Mb - Source Zip File
- [The compiled wheel files on PyPi](#)

Figure 2: BioPython library for molecular biology

```
$ sudo apt-get install python-biopython-sql
```

Sequence is the key object in bioinformatics. Sequences can be processed in BioPython with the following instructions:

```
>>> from Bio.Seq import Seq
>>> my_seq = Seq("MyDefinedSequence")
>>> my_seq
Seq('MyDefinedSequence', Alphabet())
>>> print(my_seq)
MyDefinedSequence
>>> my_seq.alphabet
Alphabet()
```

Complement and reverse complement

These are very simple — the methods return a new Seq object with the appropriate sequence and the same alphabet, as shown below:

```
>>> from Bio.Seq import Seq
>>> from Bio.Alphabet import generic_dna

>>> my_values_dna = Seq("MY_VALUES_DNA", generic_dna)
>>> my_values_dna
Seq('MY_VALUES_DNA', DNAAlphabet())

>>> my_values_dna.complement()
Seq('ATCATGTGACCA', DNAAlphabet())

>>> my_values_dna.reverse_complement()
Seq('ACCACTGTACTA', DNAAlphabet())
```

Transcription functions on DNA and RNA

If you have a DNA sequence, you may want to turn it into RNA. In bioinformatics we normally assume the DNA is the coding strand (not the template strand); so this is a simple matter of replacing all the thymines with uracil:

```
>>> my_values_dna
Seq('MY_VALUES_DNA', DNAAlphabet())

>>> my_values_dna.transcribe()
Seq('AGUACACUGGU', RNAAlphabet())
```